

Design and Implementation of a High-Throughput Experimentation Platform to Accelerate Drug Discovery

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Cite This: <https://doi.org/10.1021/acs.jmedchem.5c00814>



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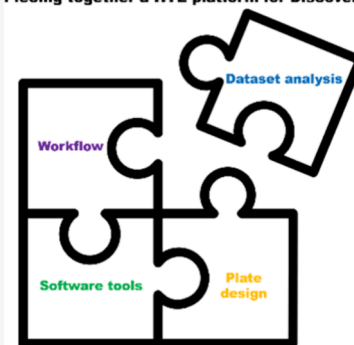
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ABSTRACT: Herein, we describe the development of the high-throughput experimentation platform and function in AbbVie's Discovery Chemistry organization. The strategy and approach tailored specifically to medicinal chemistry needs are discussed. Additionally, we disclose the top reaction conditions for the most requested transformations obtained by analysis of the combined data set generated by the platform over five years.

Piecing together a HTE platform for Discovery



SIGNIFICANCE

For HTE to be impactful in medicinal chemistry, the approach must fit the situation. To enable compound isolation and testing, HTE plate design should include a broad, diverse set of conditions to maximize the likelihood of finding a hit.

Due to time and material constraints on medicinal chemistry projects, the turnaround time must be fast and material usage must be low.

Software, which enables data quality and centralized storage, allows for a feedback loop to evaluate the approach, optimize condition sets, extract metrics and analyze combined data over time.

Analysis of our combined HTE data resulted in the top reaction condition hits for our most requested medicinal chemistry transformations (C–N coupling, C–O coupling and Suzuki coupling), which are disclosed herein.

INTRODUCTION

For decades, high throughput experimentation (HTE), the parallel evaluation of many reaction conditions for one chemical transformation, has been used to optimize routes for drug candidates in Process chemistry groups. Due to material requirements (hundreds of milligrams) and turnaround time (weeks), HTE was considered unreasonable for most synthetic challenges in medicinal chemistry. Then, about ten years ago, publications from Merck and Pfizer showed that HTE could be done with minimal amounts of material and quick turnaround time, opening the possibility of using HTE for early medicinal chemistry projects.^{1,2}

Around that time, our group sought to establish HTE capabilities for medicinal chemistry at AbbVie. As a technology group that had deep and continuous experience with parallel library synthesis for over 15 years and had a long history of enabling technologies and automation development, the group was uniquely positioned to be successful.³ Quickly, the limitations of the typical solution-based reaction assembly were obvious. Our testing discovered that for the most common transformations over 50% of commonly used reagents were not soluble in the typical reaction solvents at room temperature.

At the time, due to the challenges with automated small-scale solid dispensing, HTE plate setup was commonly done with solution-based assembly of reagents, either making fresh stock solutions before each experiment or dispensing stock solutions into well plates, drying them down and storing them cold under inert atmosphere until needed. There are many drawbacks to this approach. For example, trapped solvent remaining in the dried reagents could inhibit some reactions. Different transfer solvents had to be tested to find one that would not affect the desired reaction. Additionally, if a reagent was not soluble in the reaction solvent, experiments to find a suitable solvent to transfer a slurry accurately and reproducibly had to be performed. Inorganic bases, which are ubiquitous in

Received: March 21, 2025

Revised: October 9, 2025

Accepted: October 16, 2025



ACS Publications

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<https://doi.org/10.1021/acs.jmedchem.5c00814>
J. Med. Chem. XXXX, XXX, XXX–XXX

Table 1. Some Example Conditions in the C–N Coupling Set and Their Sources

	condition source	catalyst	ligand	base	solvent
1	top library condition	BrettPhos Pd G3		LiHMDS	dioxane
2	top library condition	Pd ₂ (dba) ₃	tetramethyl <i>t</i> -butyl XPhos	Cs ₂ CO ₃	dioxane
3	internal chemistry expert	DPPF Pd G3		Cs ₂ CO ₃	dimethoxyethane
4	internal chemistry expert	<i>tert</i> -butylXPhos Pd G3		NaOPh	toluene
5	internal chemistry expert	Pd ₂ (dba) ₃	S-BINAP	NaOtBu	toluene
6	nontraditional metals	Ni(PPh ₃) ₂ (1-naphthyl)Cl	IPr-HCl	NaOtBu	tetrahydrofuran
7	nontraditional metals	Cu(I)Cl	TMHD	Cs ₂ CO ₃	<i>N</i> -methylpyrrolidone
8	pharma publications	<i>tert</i> -butylXPhos Pd G3		DBU	<i>tert</i> -amyl alcohol

the most common metal catalyzed reactions, were particularly challenging. Research on soluble bases able to replicate the action of inorganic bases in Suzuki and C–N coupling led some groups to limit the reagents and solvents used in their HTE plates to make reaction assembly easier and create homogeneous reactions.^{4–7} Limiting the reagents and solvents that could be used was not an ideal long-term solution. Thus, our approach was to directly address the challenges associated with solid dispensing small amounts of material by inventing a way to enable automation to dispense small amounts of solid reagents reliably. ChemBeads was the result of that work.⁸ ChemBeads allowed for submilligram dispensing of solids manually or using automated solid dispensing, so that 1 μ mol scale reactions could be run using diverse reaction conditions.^{9,10} Over 400 reagent ChemBeads have been made internally over the last several years, and the technology has been embraced by other industrial and academic groups.^{11,12} Once this enabling technology was developed, our focus shifted to the three other key areas of a robust platform: screening plate design, workflow and software tools.

■ SCREENING PLATE DESIGN

When designing plates, a new approach was also taken. The industry standard at the time was a matrix approach, which can cover all combinations of the reagents included in the experiment. This approach is reasonable when you have few reaction components and/or unlimited starting material. It is also ideal when the priority is to find the best combination of reaction conditions to move forward a high value drug candidate, as in Process chemistry.

In 2017, HTE was mostly done in Process chemistry departments, which have access to larger amounts of intermediates, thus, the matrix approach was dominant. To get broad and impactful use of HTE in Discovery, it was obvious that the approach would have to be different. In Discovery, the synthetic focus is on finding a hit, a condition to enable isolation and testing of a desired compound as quickly as possible. If a compound cannot be made in less than a week, the project team will often move on to another promising and more synthetically accessible target. Thus, it was envisioned that the conditions screened should be broad and diverse to maximize the chance of finding a hit. A matrix approach that included all the diverse conditions for a transformation, like C–N coupling, would result in an extremely large number of reactions that would not be feasible to run and analyze with medicinal chemistry project material and time limits.

Therefore, our strategy was to develop a diverse, curated set of conditions from many sources. Critically, our invention of ChemBeads allowed diverse conditions to be used. No limitations on solids or solvents were set when collating the set of conditions. First, for the C–N coupling set, the most

robust and general conditions from years of parallel library experience, such as Table 1, entries 1 and 2, were compiled.³ The “go-to” conditions from internal medicinal chemistry experts and contract research organizations were added (Table 1, entries 3–5); many of these were unpublished condition combinations at the time.¹³ Published methods with nontraditional metals, that might show differentiating reactivity were included (Table 1, entries 6–7).^{14,15} Publications from pharmaceutical companies, especially containing conditions used robustly on large scale, were selected as an area of focus as the substrates were more relevant than the simple substrates used in the initial publications of novel methods. For example, the selective coupling of 1-methyl-1H-pyrazol-4-amine hydrochloride with an unprotected bromo-pyrrolopyrimidine as reported by Tao et al. at Pfizer led us to add this condition to our set (Table 1, entry 8).¹⁶

The compiled sets were reviewed by several medicinal chemists at AbbVie for feedback and modification. As the sets were used, they were edited over the years, and now, an average of 70 conditions are tested per experiment. Conditions that never worked were reevaluated to make sure their published scopes were reproducible. Newly published conditions or ones identified internally for difficult reactions were added. For example, a medicinal chemist suggested a condition that had worked for the coupling of *tert*-butyl carbamate with a sensitive aryl bromide at room temperature (Table 1, entry 4). The use of the weak phenoxide base at room temperature allowed for better functional group tolerance than using sodium *tert*-butoxide at 80 °C or higher, which is a very common C–N coupling condition.¹⁷ Another suggestion from a chemist was a condition identified to work uniquely well after a C–N coupling HTE failed to find a hit (Table 1, entry 5).¹⁸ This condition was added to evaluate it for complementarity to other conditions in the set. In 2020, we had a short consultation collaboration with John Hartwig to evaluate and suggest enhancements to our C–N coupling set. A few conditions were added that year that became hits for 10% of the screens done from 2021 to 2023.

As we expanded the list of HTE transformations, a matrix approach for plate design became relevant for some cases. For example, unlike C–N coupling, which had many years of research and literature history when the set was compiled, nickel-catalyzed photoredox transformations were added when only a few publications existed, and internal experience was limited. A diverse, curated set could not be compiled in the same way, so a matrix approach was taken. Over time, with additional publications and extensive internal validation, these sets were reevaluated and edited to add conditions in a more deliberate, curated way.

While sets for the most frequently used metal catalyzed medicinal chemistry transformations, C–N coupling, Suzuki

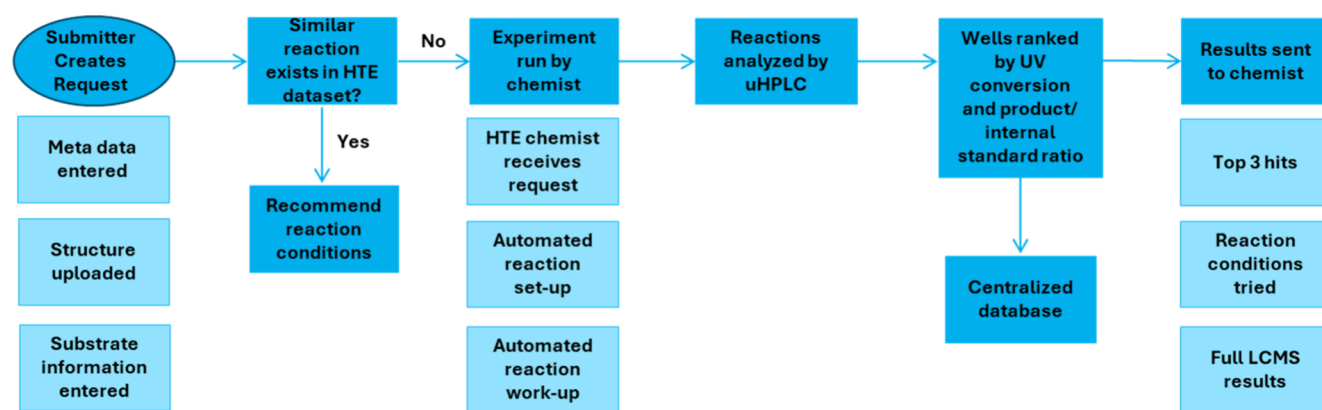


Figure 1. Standard HTE workflow.

coupling and C–O coupling, were developed first, quickly we expanded into newer areas. Demand for more sp^2/sp^3 cross coupling methods and late-stage functionalization led us to develop several sets in each area. Additionally, demand for other transformations by project need led to the development of many sets in the first few years. Over time, developing new sets became less of a priority, but evaluating and updating existing transformations continues to be important.

WORKFLOW

When designing the workflow, the focus was on getting a hit with an average 2-day turnaround time and a 70% success rate; success was defined as identifying at least one condition that could enable isolation of the desired product. Automated solid (ChemSpeed) and liquid dispensing (Tecan Fluent) were crucial to meeting this goal. Typically, experiments are run using approximately 1 mg of limiting reagent per reaction, although using 0.25 mg is possible. The main limitations on scale are small solvent volumes and analytical instrumentation. The standard workflow is initiated by a request from a chemist in our dedicated HTE web application (Figure 1). The chemist can enter structures, record meta data, list reaction conditions tried and enter any relevant substrate information, like solubility or stability issues. Next, a HTE chemist runs the desired transformation using automated reaction setup and workup protocols. For reagents, automated solid dispensing is used to rapidly assemble plates. For the reaction partners, manual dispensing of ChemBeads using scoops is done to use a minimum amount of material. Manual dispensing of reagents is occasionally done when assembling custom plates for a new transformation; the process is very slow and prone to error. After the reaction is complete, 4,4'-di-*tert*-butylbiphenyl is added as an internal standard. uHPLC-MS data is collected and analyzed; the reaction conditions are ranked by product/internal standard ratio and UV conversion. DESI-MS was evaluated to triage successful reactions and shorten the uHPLC-MS queue. The fragmentation of complex molecules, often with protecting groups, during ionization led to false negatives when only looking for the desired mass. Thus, it is not used routinely. Reaction results are sent to the chemist and to a database, which allows for further analysis.

When communicating results, three of the top conditions (when possible) are given along with the LCMS traces and a list of conditions tried. Three conditions are given to facilitate short turnaround times from HTE results to isolation of the desired product. For example, if the reagents for the best

condition are not immediately available, another condition that gives acceptable results could be used. Additionally, if separation issues or an undesirable byproduct is formed with the top condition, alternatives are immediately available. When submitting a HTE request, one test reaction must have already been done to minimize unnecessary submissions, which waste resources and elongate the queue.

In the past few years, another step has been added to the workflow. A suggestion is often given for the best condition(s) to try before a screen is run; the suggestion is based on historical data generated using similar structures. This has given rapid access to enabling conditions without HTE and has further decreased the number of unnecessary submissions.

SOFTWARE TOOLS

In contrast with how HTE data was often stored at the time, in single excel files on individual computers or attached to chemistry electronic notebook (ChemELN) pages, a centralized database for all experiments was set up within months of initiating HTE in Discovery. Even at this early stage, the power of future data set analysis was obvious. Long-term planning for streamlined data analysis and storage was a priority. A software tool, the HTE LCMS Browser, was developed while the evaluation and implementation of commercial HTE software was undertaken (Figure 2).

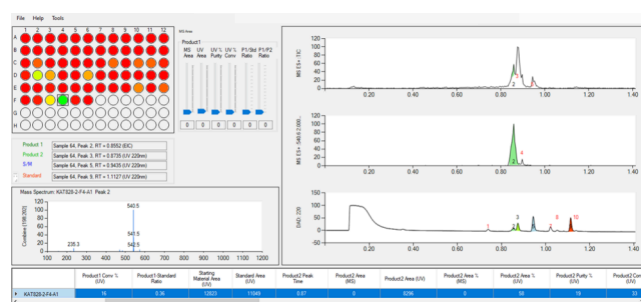


Figure 2. HTE LCMS browser user interface.

The “HTE LCMS Browser” was able to load in report (rpt) files saved on a network drive by the Waters uHPLC-MS instrument. A 96-well plate of the data in the top left is linked to the LCMS data displayed on the right, so that choosing a well displays the data associated with it. Peaks could be selected as the starting material, product, internal standard and secondary products/byproducts. The selection propagates

across the whole plate, so any peak at the same retention time is marked. This allows for rapid calculation of starting material conversion and product/internal standard (P/IS) ratio and quick ranking of reactions across the plate using a heat map feature. Manual sliders allow a user to quickly filter reactions above a threshold for MS Area, UV Area, UV % purity, UV % conversion, P/IS ratio or product 1/product 2 ratio. In addition, this tool can generate and save a comma-separated values (csv) file containing the data to a centralized database, where it could be queried and viewed in aggregate with data from other experiments using Spotfire by TIBCO. Even though the HTE LCMS Browser was very powerful, there were key benefits to pursuing an all-in-one software tool, such as storing all the relevant meta data with the reaction details in a ChemELN and the ability to reprocess the raw data, which can increase data quality.

After evaluating commercial options in 2019, ACD/Lab's Katalyst D2D was the most advanced software available for HTE. After an initial pilot showed that the software could meet the current workflow needs with some customization, capital budget support was received to purchase the software. Many of users' favorite features from the HTE LCMS Browser, described above, were added. With this software in place, any chemist at the company can submit a request through the Katalyst D2D web application. A HTE chemist accepts the request, custom designs a reaction plate or chooses a reaction template for the reaction and analysis plates, creates a sequence file, which is sent to the uHPLC, and analyzes the data after it is collected by the instrument and automatically swept into the database. In collaboration with ACD/Laboratories, tremendous work went into integrating the software with our instrumentation to ensure the data can flow back and forth quickly and easily. Additionally, considerable effort went into building a custom view of the database to retrieve the data and combine it with the legacy HTE data set originating from the HTE LCMS browser.

This large, comprehensive data set allowed HTE metrics tracking from 2017 to present and analysis of success rates and trends in the reaction conditions and transformations (Table S1). The ability to access and control this data set is one reason HTE was never democratized; only HTE experts run the screens. This data set has been used to optimize the sets over time, but this feedback loop has also allowed evaluation of our overall approach. One important metric is adoption. Since mid-2017, over 100 medicinal chemistry programs have been impacted by the Discovery HTE platform. Over 150 medicinal chemists have submitted at least one request, and over 30 chemists have submitted 5 or more requests. On average over the last seven years, 30 projects and 46 chemists have submitted at least one request each year. This highlights the wide acceptance and utilization of the platform by the chemistry community; submitting requests is routine for projects when they encounter challenging chemistry.

DATA SET ANALYSIS

In total, over 800 screens were completed by 2023, covering 20 transformations (Table 2). The most often requested transformations were 313 requests for C–N coupling, 208 for Suzuki coupling and 77 for Ullmann type C–O coupling. These numbers reflect the dominant use of Suzuki and C–N coupling over other metal catalyzed transformations in medicinal chemistry, which has been well documented by previous analyses.^{19–21,3,22} These requests also highlight that

Table 2. Number of Experiments for Each Transformation

transformation	#
amide coupling	9
C–N coupling	313
C–O coupling	77
reductive cross-electrophile coupling	3
Heck coupling	5
late-stage fluorination	25
late-stage methylation	19
late-stage oxidation (chemical, biocatalytic)	39
late-stage trifluoromethylation	1
photoredox Copper sp ³ C–N coupling	4
photoredox Nickel BF ₃ K coupling	31
photoredox Nickel cross-electrophile coupling	5
photoredox Nickel C–N coupling	2
photoredox Nickel decarboxylative coupling	39
photoredox Nickel HAT C–H activation	1
ring closing metathesis	4
aromatic nucleophilic substitution	2
Sonogashira coupling	13
sulfonamide formation	1
Suzuki coupling	208
total	801

despite decades of development, there are still unmet needs for these two critical transformations for drug discovery. Additionally, the large number of substrate dependent variables and published conditions and ligands make it hard to pick the best condition for a C–N coupling reaction, which leads to the dominance of this transformation. In our experience, if Suzuki coupling conditions are identified that will work for a given (hetero)aryl bromide and boronic acid, the conditions will often translate to couple the same aryl bromide with many related boronic acids, although the yields may vary. If the product can be isolated, finding a better condition is often unnecessary. However, for C–N coupling, rescreening conditions for the same scaffold with a new amine of interest is common and often necessary.

In the subsection of photoredox transformations, the nickel catalyzed photoredox decarboxylative coupling is the most requested, followed closely by the BF₃K salt coupling. The decarboxylative coupling can benefit greatly from HTE.⁹ There are several variables that can be optimized, including additives, which can have a large impact on conversion.²² In contrast, the nickel catalyzed photoredox deoxygenative coupling is not offered as an HTE transformation because it often works robustly using the published conditions and many variables that impact the reaction outcome have not yet been identified.²³ In the subsection of late-stage functionalization (LSF), oxidation has the highest number of experiments, but it also had the advantage of being the first LSF transformation offered. Additionally, it includes chemical oxidation and biocatalytic oxidation. It is not uncommon to screen a substrate for both as they can show complementary reactivity.

Late-stage fluorination is a highly desirable transformation that was investigated internally, but for many years, the field was not advanced enough to be reliably used on complex medicinal chemistry scaffolds. In 2018, when validating literature methods, the substrate scope rarely extended beyond what was published. Late-stage fluorination was finally established in 2023 as an available transformation after a more promiscuous and reliable set of conditions was identified.

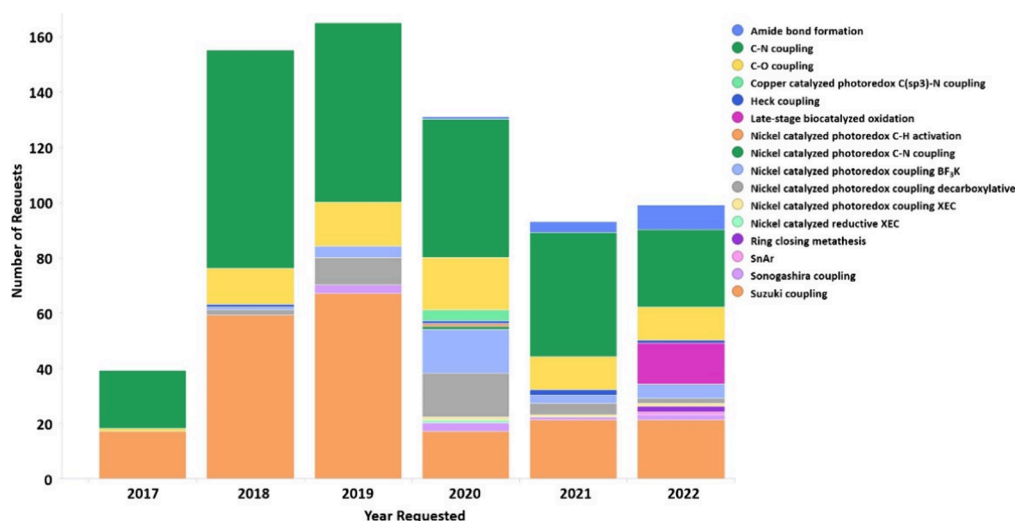
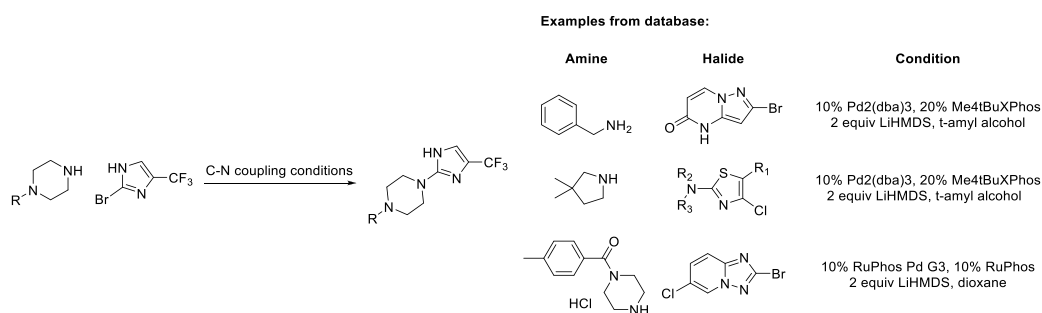


Figure 3. Number of experiments for each transformation by year.



Reaction	Catalyst	Ligand	Base	Solvent	Result
1	BrettPhos Pd G3	BrettPhos	CS ₂ CO ₃	Dioxane	No desired product
2	Pd ₂ (dba) ₃	XPhos	NaOtBu	DME	No desired product
3	RuPhos Pd G3	RuPhos	CS ₂ CO ₃	Dioxane	No desired product
4	BrettPhos Pd G3	BrettPhos	LiHMDS (1M in THF)	Dioxane	No desired product
5	Pd ₂ (dba) ₃	Me ₄ tBuXPhos	LiHMDS (1M in THF)	<i>t</i> -amyl OH	45% conversion
6	RuPhos Pd G3	RuPhos	LiHMDS (1M in THF)	THF	No desired product

Figure 4. Substructure searching of historical HTE data to recommend best conditions.

Although it was established after late-stage methylation, it quickly became requested more frequently. It is expected that the number of late-stage functionalization experiments will only continue to grow over the next few years.

In 2018, a searchable database of HTE reactions was set up and made available for HTE chemists to search; substructure search was used to find similar reaction partners, and the top conditions that were identified in the experiment. As more experiments were accumulated, this addition to the workflow starting in late 2020 proved to be valuable and eliminated the need for some screens to be run. In lieu of an experiment, relevant reaction conditions were recommended. Thus, the number of requests per year has declined from a peak of over 160/year to stabilize around 100/year (Figure 3).

Additionally, in cases where material was so limited that HTE was not possible, recommendations enable desired compounds to still be accessed (Figure 4). In one instance, a chemist had less than 200 mg of a precious secondary amine and had already tried commonly used C–N coupling conditions with no success (Figure 4, entries 1–4). The chemist did not want to use half of their material on a HTE campaign. Since the heterocycle was deemed to be the difficult coupling partner, a search of the HTE database focused on related five-membered heterocycles. Two conditions, which had worked for similar coupling partners, were identified in the data set. Both conditions were tried, one gave 45% conversion to product (Figure 4, entry 5). These conditions were used to

enable parallel library synthesis with a small, focused set of heteroaryl bromides.

To make this valuable information readily available to all AbbVie chemists, an internal tool was developed in 2023 to recommend conditions based on substructure search. Similar to frequently used commercial tools that search literature and patent databases, the tool searches our internal HTE data set and ChemELN records. Combining these two very relevant databases into one searchable tool has been incredibly enabling.

As noted above, one of the most commonly requested HTE transformations is the Suzuki reaction. Across all the Suzuki HTE plates, a 78% success rate has been achieved (Table S1). The success rate is the percentage of experiments which identify at least one viable hit, usually over 10% conversion to desired product. The success rate has increased from 77% from 2017 to 2019 (first 3 years) to 82% from 2020 to 2022 (next 3 years). This increase is attributed to the conditions added to the set. In 38% of Suzuki screens run during 2020–2023, conditions added after 2019 were top 3 hits. In two experiments, the only hits were from these newly added conditions. This highlights the importance of rapidly evaluating and incorporating newly published or internally identified conditions to increase success rates.

The most successful conditions are not diverse when the results from all the Suzuki experiments are aggregated. Several catalyst systems repeat in the top rankings with different bases (Table 3). For example, PdCl₂(dtbpf) appears three times;

Table 3. Conditions Most Frequently Found as Top Hits in Suzuki Coupling HTE Reactions

rank	catalyst	base	solvent	total top 1 hits	total top 3 hits
1	PdCl ₂ (dtbpf)	K ₃ PO ₄	dioxane/H ₂ O (4:1)	10	24
2	P(tBu) ₃ Pd G3	K ₃ PO ₄	dioxane/H ₂ O (4:1)	12	20
3	XPhos Pd G3	K ₃ PO ₄	dioxane/H ₂ O (4:1)	9	20
4	PdCl ₂ (dtbpf)	Cs ₂ CO ₃	dioxane/H ₂ O (4:1)	7	18
5	CataCXium A Pd G3	Cs ₂ CO ₃	dioxane/H ₂ O (4:1)	5	18
6	Pd(Amphos) ₂ Cl ₂	K ₂ CO ₃	dioxane/H ₂ O (4:1)	4	17
7	PdCl ₂ (PPh ₃) ₂	Cs ₂ CO ₃	dioxane/H ₂ O (4:1)	8	15
8	PdCl ₂ (dtbpf)	K ₂ CO ₃	dioxane/H ₂ O (4:1)	2	15
9	Pd(dppf)Cl ₂ – CH ₂ Cl ₂	K ₂ CO ₃	dioxane/H ₂ O (4:1)	6	14
10	Pd(dppf)Cl ₂ – CH ₂ Cl ₂	Cs ₂ CO ₃	dioxane/H ₂ O (4:1)	3	14

Pd(Amphos)₂Cl₂, Pd(dppf)Cl₂–CH₂Cl₂, Pd(DPEPhos)Cl₂ and XPhos Pd G3 appear two times each. An analysis of the top conditions by halide substructure shows that many of the top conditions for each substructure are represented in the overall top conditions (Table S2). A similar result is achieved by analysis of the top conditions by boronate substructure (Table S3). This indicates that for most Suzuki requests, a small, focused screen would be sufficient to find conditions for medicinal chemists to scale-up and isolate the desired product.

As noted above, the most requested HTE transformation is C–N coupling, where a 67% overall success rate has been

achieved. Interestingly, the success rate stayed steady at 67% for the first three years from 2017 to 2019 and in the second three years from 2020 to 2022, even though new conditions were added. When looking at individual experiments, however, the addition of new conditions seems to have had a profound impact on the outcome. For 26% of experiments from 2021 to 2023, the top condition had been added to the set after 2020. The trend holds when looking at the top three conditions for all experiments during this time frame; 27% of the hits were conditions added after 2020. This highlights the importance of continually and frequently updating the screening set. In almost 10% of C–N coupling experiments, only new conditions were identified as hits. If these conditions were not added, those experiments would have failed, and the targets may have been deprioritized.

When looking at the aggregated data, five variations of conditions using BrettPhos Pd G3 appear in the top rankings (Table 4). Additionally, LiHMDS appears three times near the top of the list; the frequent appearance of LiHMDS was also a casual observation made over the years. This base seems to perform particularly well with the heterocycles commonly seen in medicinal chemistry programs, especially five membered heterocyclic halides (Table S4). When doing the substructure analysis of the top C–N coupling conditions, the overall top conditions are well represented, but there are several conditions that are not. This reflects the more complicated nature of identifying C–N coupling conditions. Both partners can have a strong effect on the reaction conditions necessary for success.

The C–O coupling reaction has a much lower success rate at 55%. Additionally, the success rate dropped from 60% in the first three years (2017–2019) to 51% in the next three years (2020–2022). For the C–O coupling, only one experiment identified a condition that was added after 2020. Thus, the newer conditions that have been added have not improved the success rate over time (Table 5). Even though interest and publications on C–O coupling have had a resurgence, the newly published conditions have not been superior with structures relevant to medicinal chemists. One reason may be the type of requests received for this transformation. For the C–O coupling set, there are two types of reactions that are often submitted. The first are alkyl alcohol couplings. There has been a large increase in requests of this type, and despite recent literature reports, these substrates have still been a difficult class that often fail.^{24–28}

The second type of submissions are amine couplings that failed or only showed conversion with copper catalysts using our C–N coupling plate. These reactions can be rescreened using our copper dominated C–O coupling plate, especially if one of the coupling partners has functional groups that could inhibit palladium catalysis. These requests are inherently difficult since they have already failed or given a low conversion with the C–N coupling plate. When retrospectively analyzing transformations that have been run in both the C–O and C–N coupling sets, it is obvious that some examples should not have been run because there was not a strong rationale. Experiments that identified a copper catalyst as the top hit in the C–N coupling set are the only examples that benefited from being run in the C–O coupling set. These observations will help guide future submissions.

Table 4. Conditions Most Frequently Found as Top Hits in C–N Coupling HTE Reactions

rank	catalyst	ligand	base	solvent	total top 1 hits	total top 3 hits
1	RuPhos Pd G3	RuPhos	LiHMDS	dioxane	10	18
2	BrettPhos Pd G3	BrettPhos	CS ₂ CO ₃	dioxane	10	15
3	BrettPhos Pd G3	BrettPhos	LiHMDS	dioxane	8	15
4	BrettPhos Pd G3		CS ₂ CO ₃	dioxane	5	15
5	Pd ₂ (dba) ₃	Me ₄ tBuXPhos	LiHMDS	<i>t</i> -amylOH	7	14
6	BrettPhos Pd G3	RuPhos	K ₃ PO ₄	<i>t</i> -amylOH/DMA	4	11
7	BrettPhos Pd G3	BrettPhos	K ₃ PO ₄	<i>t</i> -amylOH	2	10
8	BINAP Pd G3		CS ₂ CO ₃	dioxane	6	9
9	RuPhos Pd G3		CS ₂ CO ₃	dioxane	3	9
10	XPhos Pd G3		CS ₂ CO ₃	dioxane	5	8

Table 5. Conditions Most Often Found as Top Hits in C–O Coupling HTE Reactions

	catalyst	ligand	base	solvent	total top 1 hits	total top 3 hits
1	RockPhos Pd G3		CS ₂ CO ₃	toluene	3	8
2	Pd(OAc) ₂	BippyPhos	CS ₂ CO ₃	toluene	4	6
3	<i>t</i> BuXPhos Pd G3		CS ₂ CO ₃	toluene	2	4
4	JosiPhos Pd G3		CS ₂ CO ₃	toluene	2	4
5	CuI	trans-1,2-cyclohexane-dimethanamine	KOtBu	acetonitrile	3	3
6	CuI	2,6-dimethoxyanilino(oxo)acetic acid	K ₃ PO ₄	DMSO	2	3
7	CuI	1,10-phenanthroline	CS ₂ CO ₃	toluene	1	3
8	CuI	L-Proline	CS ₂ CO ₃	DMF	1	3
9	Pd ₂ (dba) ₃	CataCXium PIntB	CS ₂ CO ₃	toluene	1	3
10	Cu ₂ O	salicylaldehyde	CS ₂ CO ₃	acetonitrile	2	2

■ UNMET NEEDS

By reviewing failed experiments over the years, the remaining gaps in the substrate scope of each transformation appear. For C–O coupling, as mentioned above, alkyl alcohols can still be challenging, especially when paired with five-membered heterocyclic bromides. In particular, tertiary alkyl alcohols are an unsolved problem. Sensitive functional groups, such as dihydrouridines, which have become more common with the rise in PROTAC programs, are prone to solubility issues and ring-opening, and are not yet common substrates demonstrated in new method development publications. With its low success rate, the C–O coupling set could be a focus for improvement. Expanding the set of conditions with more solvents and additives is one option. Another is to look at the limitations of our platform. For some of the reactions that failed in the C–O set, literature searches indicate that higher reaction temperatures (up to 130 °C), may have allowed them to work. Due to limitations in our workflow, only temperatures of up to 110 °C have been used previously. To increase the success rate of these requests going forward, reevaluation of our workflow and reactors is a priority.

For C–N coupling, sterically hindered amines, especially those with bis-substitution alpha to the amine, are challenging. Amides and ureas, which can be chelators for palladium, are especially troublesome if they are on a heterocyclic halide. Even for the Suzuki coupling, which is perceived to be very robust, some limitations exist. Electron poor boronic acids and sterically hindered halides are difficult partners to couple. Certain five membered heteroaryl bromides, such as thiophenes, thiazoles and triazoles, can be difficult as well. While alkyl boronic acids are generally not stable, their potassium trifluoroborate salt counterparts can be successful in Suzuki or photoredox coupling reactions.

■ FUTURE OUTLOOK

Initially, our HTE development focused on technology, screening plate design, workflow and software. With our approach to screening plate design, a shift from 96 to 384 or 1536 reactions has not been a priority. More experiments may not translate into a higher chance of finding a hit. The implementation of 384 well plates is interesting from a technological aspect and may be useful for building data sets. However, we have not seen the need to implement it routinely in our HTE workflow.

Our HTE development now focuses on data analytics. As data accumulates, it is important to consider how it can be used. As mentioned above, our data is already being used to replace screens with condition recommendations. The success of this workflow addition has expanded into the development of a reaction condition recommendation tool, which will allow for structure inputs. Concurrently, the use of Bayesian Optimization (BO) and machine learning models are also decreasing the number of necessary experiments, and this trend is expected to continue in the future.^{6,29–35}

A future area of development critical to HTE and training machine learning models is reaction quantitation. While analytical methods have gotten faster to address an increase in reactions, this speed can come at the cost of data quality. Mass spectrometry methods, such as DESI-MS or Echo-MS, can analyze a whole plate in minutes but are only appropriate to triage reactions.^{36,37} Ultrahigh-performance liquid chromatography mass spectrometry (uHPLC-MS) is still the dominant tool. Ignoring product solubility or ionization issues, the industry standard of product quantitation based on product UV area has problems. Overlapping UV peaks can complicate calculations. Additionally, comparing within a plate to find the best reaction can be done, but making comparisons between plates, different reactions or reactions that used different analytical methods is not advisable. Many groups have

investigated and compared ELSD and CAD as other ways to evaluate reaction success with enthusiasts and critics on each side. No rigorous comparison of diluents and internal standard combinations or a comparison of analytical methods, techniques or instrumentation for HTE can be found in the literature, which, considering the resources it would take, is understandable. As this set of challenges is of interest to many in the field, collaboration through a precompetitive consortium or with an academic group to share data sets or experimental support could be a way to accelerate the development of reaction quantitation, which would benefit all. A collaborative solution to reaction quantitation could also lead to a consensus in the field on the best reaction output for machine learning.

■ ASSOCIATED CONTENT

SI Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.jmedchem.5c00814>.

Data tables of success rates and top condition analysis not included in main text (PDF)

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Author Contributions

The manuscript was written through contributions of all authors. All authors have given approval to the final version of the manuscript.

Notes

The authors declare no competing financial interest.

Biographies

Amanda W. Dombrowski is a Principal Research Scientist at AbbVie. She received her B.A. (2007) in Chemistry from Goucher College and her Ph.D. (2012) in Organic Chemistry from Boston College with Kian Tan. Her postdoctoral research (2012) was done with Samuel Stupp at Northwestern University. Since joining AbbVie and the Advanced Chemistry Technologies (ACT) group in 2015, she has investigated methods for parallel synthesis, helped develop the technology for and implemented the high-throughput experimentation platform, led data analysis projects, as well as contributed to several medicinal chemistry programs.

Ana L. Aguirre, born and raised in El Paso, TX, pursued her education at the University of Texas at El Paso, earning a B.Sc. in Chemistry in 2004 and an M.Sc. in Chemistry in 2006. She has been a member of the Advanced Chemistry Technologies (ACT) team at AbbVie since 2006, specializing in library synthesis, Pd-mediated cross-couplings, method development, and high-throughput experimentation.

Ying Wang received her Ph.D. in Organic Chemistry from Iowa State University and completed a Postdoctoral Fellowship at the University of Chicago prior to starting her career in industry in 2003. Ying has led multiple initiatives at AbbVie and has been involved in several medicinal chemistry programs. Since 2014, she has headed up the Advanced Chemistry Technologies (ACT) group at AbbVie, a centralized chemistry technology group that specializes in parallel library synthesis, high-throughput experimentation, reaction informatics, and the investigation and implementation of various chemistry technologies for drug discovery research.

■ ACKNOWLEDGMENTS

We thank the following AbbVie scientists: Noah Tu for his automation expertise and ongoing ChemBead innovations, which have enabled this work, Philip Searle for the development of the HTE LCMS Browser software, and Brian Wakefield for setting up our searchable database of HTE reactions and making it easily accessible to chemists.

■ ABBREVIATIONS USED

BO, Bayesian optimization; BINAP Pd G3, [2'-(amino-κN)[1,1'-biphenyl]-2-yl-κC][[2'-(diphenylphosphino)[1,1'-binaphthalen]-2-yl]diphenylphosphine-κP](methanesulfonato-κO)-palladium; BippyPhos, 5-(*di**tert*-butylphosphino)-1', 3', 5'-triphenyl-1'*H*-[1,4']bipyrazole; BrettPhos, 2-(dicyclohexylphosphino)3,6-dimethoxy-2',4',6'-triisopropyl-1,1'-biphenyl; BrettPhos Pd G3, [(2-dicyclohexylphosphino)3,6-dimethoxy-2',4',6'-triisopropyl-1,1'-biphenyl]-2-(2'-amino-1,1'-biphenyl)]palladium(II) methanesulfonate; methanesulfonate; CAD, charged aerosol detector; CataCXium A Pd G3, [(di(1-adamantyl)-butylphosphine)-2-(2'-amino-1,1'-biphenyl)]palladium(II) methanesulfonate; CataCXium PIntB, 2-(*di**tert*-butylphosphino)-1-phenylindole; ChemELN, chemistry electronic notebook; Cs₂CO₃, cesium carbonate; Cu₂O, copper(I) oxide; CuI, copper(I) iodide; DBU, 1,8-diazabicyclo[5.4.0]undec-7-ene; DESI-MS, desorption electrospray ionization-mass spectrometry; DMA, *N,N*-dimethylacetamide; DMF, *N,N*-dimethylformamide; DME, 1,2-dimethoxyethane; DPPF Pd G3, methanesulfonato 1,1'-ferrocenediylbis(diphenylphosphino) (2'-amino-1,1'-biphenyl-2-yl) palladium(II); Echo-MS, acoustic ejection mass spectrometry; ELSD, evaporative light scattering detector; H₂O, water; HTE, high-throughput experimentation; *i*Pr-HCl, *N,N'*-(2,6-diisopropylphenyl)dihydroimidazolium chloride; JosiPhos Pd G3, Josiphos SL-J009-1 Pd G3, {(R)-1-[(Sp)-2-(dicyclohexylphosphino)ferrocenyl]ethyl-di-*tert*-butylphosphine}[2-(2'-amino-1,1'-biphenyl)]palladium(II) methanesulfonate; K₂CO₃, potassium carbonate; K₃PO₄, potassium phosphate tribasic; KOtBu, potassium *tert*-butoxide; LCMS, liquid chromatography-mass spectrometry; LiHMDS, lithium bis(trimethylsilyl)amide; LSF, late-stage functionalization; NaOPh, sodium phenoxide; NaOtBu, sodium *tert*-butoxide; Ni(PPh₃)₂(1-naphthyl)Cl, *trans*-Chloro(1-naphthyl)bis(triphenylphosphine)nickel(II); NMP, *N*-methylpyrrolidone; Pd₂(dba)₃, tris(dibenzylideneacetone)-dipalladium(0); Pd(Amphos)2Cl₂, bis(*di**tert*-butyl(4-dimethylaminophenyl)phosphine)dichloropalladium(II); PdCl₂(dtbpf), [1,1'-bis(*di**tert*-butylphosphino)ferrocene]-dichloropalladium(II); PdCl₂(PPh₃)₂, bis-(triphenylphosphine)palladium(II) dichloride; Pd(DPEPhos)-Cl₂, dichloro[bis(2-(diphenylphosphino)phenyl)ether]-palladium(II); Pd(dppf)Cl₂-CH₂Cl₂, [1,1'-

bis(diphenylphosphino)ferrocene]dichloropalladium(II), complex with dichloromethane; Pd(OAc)₂, palladium(II) acetate; PROTAC, proteolysis-targeting chimera; P(tBu)₃ Pd G3, [2'-(amino-κN)[1,1'-biphenyl]-2-yl-κC](methanesulfonato-κO)-[tris(1,1-dimethylethyl)phosphine]palladium; RockPhos Pd G3, [(2-ditert-butylphosphino-3-methoxy-6-methyl-2',4',6'-triisopropyl-1,1'-biphenyl)-2-(2-aminobiphenyl)]palladium(II) methanesulfonate; RuPhos, 2-dicyclohexylphosphino-2',6'-diisopropoxybiphenyl; RuPhos Pd G3, (2-dicyclohexylphosphino-2',6'-diisopropoxy-1,1'-biphenyl)[2-(2'-amino-1,1'-biphenyl)]palladium(II) methanesulfonate; S-BINAP, (S)-(-)-2,2'-bis(diphenylphosphino)-1,1'-binaphthalene; t-amyl-OH, *tert*-amyl alcohol; THF, tetrahydrofuran; TMHD, 2,2',6,6-tetramethylheptane-3,5-dione; *tert*-butylXPhos Pd G3, [(2-ditert-butylphosphino-2',4',6'-triisopropyl-1,1'-biphenyl)-2-(2'-amino-1,1'-biphenyl)] palladium(II) methanesulfonate; tetramethyl tBuXPhos, 2-ditert-butylphosphino-3,4,5,6-tetramethyl-2',4',6'-triisopropyl-1,1'-biphenyl; uHPLC, ultrahigh-performance liquid chromatography; UV, ultraviolet; XPhos, 2-dicyclohexylphosphino-2',4',6'-triisopropylbiphenyl; XPhos Pd G3, (2-dicyclohexylphosphino-2',4',6'-triisopropyl-1,1'-biphenyl)[2-(2'-amino-1,1'-biphenyl)]palladium(II) methanesulfonate

REFERENCES

- (1) Buitrago Santanilla, A.; Regalado, E. L.; Pereira, T.; Shevlin, M.; Bateman, K.; Campeau, L.-C.; Schneeweis, J.; Beritt, S.; Shi, Z.-C.; Nantermet, P.; Liu, Y.; Helmy, R.; Welch, C. J.; Vachal, P.; Davies, I. W.; Cernak, T.; Dreher, S. D. Nanomole-Scale High-Throughput Chemistry for the Synthesis of Complex Molecules. *Science* **2015**, *347* (6217), 49–53.
- (2) Perera, D.; Tucker, J. W.; Brahmabhatt, S.; Helal, C. J.; Chong, A.; Farrell, W.; Richardson, P.; Sach, N. W. A Platform for Automated Nanomole-Scale Reaction Screening and Micromole-Scale Synthesis in Flow. *Science* **2018**, *359* (6374), 429–434.
- (3) Dombrowski, A. W.; Aguirre, A. L.; Shrestha, A.; Sarris, K. A.; Wang, Y. The Chosen Few: Parallel Library Reaction Methodologies for Drug Discovery. *J. Org. Chem.* **2022**, *87*, 1880–1897.
- (4) Zickermann, V.; Wirth, C.; Nasiri, H.; Siegmund, K.; Schwalbe, H.; Hunte, C.; Brandt, U. Mechanistic Insight from the Crystal Structure of Mitochondrial Complex I. *Science* **2015**, *347* (6217), 44–49.
- (5) Buitrago Santanilla, A.; Christensen, M.; Campeau, L. C.; Davies, I. W.; Dreher, S. D. P2Et Phosphazene: A Mild, Functional Group Tolerant Base for Soluble, Room Temperature Pd-Catalyzed C-N, C-O, and C-C Cross-Coupling Reactions. *Org. Lett.* **2015**, *17* (13), 3370–3373.
- (6) Reizman, B. J.; Wang, Y. M.; Buchwald, S. L.; Jensen, K. F. Suzuki-Miyaura Cross-Coupling Optimization Enabled by Automated Feedback. *React. Chem. Eng.* **2016**, *1* (6), 658–666.
- (7) Dennis, J. M.; White, N. A.; Liu, R. Y.; Buchwald, S. L. Breaking the Base Barrier: An Electron-Deficient Palladium Catalyst Enables the Use of a Common Soluble Base in C-N Coupling. *J. Am. Chem. Soc.* **2018**, *140* (13), 4721–4725.
- (8) Tu, N. P.; Dombrowski, A. W.; Goshu, G. M.; Vasudevan, A.; Djuric, S. W.; Wang, Y. High-Throughput Reaction Screening with Nanomoles of Solid Reagents Coated on Glass Beads. *Angew. Chem.* **2019**, *131* (24), 8071–8075.
- (9) Impastato, A. C.; Brown, J. T. C.; Wang, Y.; Tu, N. P. Readily Accessible High-Throughput Experimentation: A General Protocol for the Preparation of ChemBeads and EnzyBeads. *ACS Med. Chem. Lett.* **2023**, *14* (4), 514–520.
- (10) Gesmundo, N. J.; Tu, N. P.; Sarris, K. A.; Wang, Y. ChemBeads-Enabled Photoredox High-Throughput Experimentation Platform to Improve C(Sp²)-C(Sp³) Decarboxylative Couplings. *ACS Med. Chem. Lett.* **2023**, *14* (4), 521–529.
- (11) Glogowski, M. P.; Cercizi, N.; Lynch-Colameta, T.; Ridgers, L. H.; Phelan, J. P.; Rowley, A. M.; Rauch, M. P. Utilization of High-Throughput Experimentation (HTE) and ChemBeads Toward the Development of an Aryl Bromide and Benzyl Bromide Photoredox Cross-Electrophile Coupling. *Org. Lett.* **2024**, *26* (12), 2420–2424.
- (12) Aguirre, A. L.; Loud, N. L.; Johnson, K. A.; Weix, D. J.; Wang, Y. ChemBead Enabled High-Throughput Cross-Electrophile Coupling Reveals a New Complementary Ligand. *Chem. - Eur. J.* **2021**, *27* (51), 12981–12986.
- (13) Robbins, D. W.; Noviski, M. A.; Tan, Y. S.; Konst, Z. A.; Kelly, A.; Auger, P.; Brathaban, N.; Cass, R.; Chan, M. L.; Cherala, G.; Clifton, M. C.; Gajewski, S.; Ingallinera, T. G.; Karr, D.; Kato, D.; Ma, J.; McKinnell, J.; McIntosh, J.; Mihalic, J.; Murphy, B.; Panga, J. R.; Peng, G.; Powers, J.; Perez, L.; Rountree, R.; Tenn-McClellan, A.; Sands, A. T.; Weiss, D. R.; Wu, J.; Ye, J.; Guiducci, C.; Hansen, G.; Cohen, F. Discovery and Preclinical Pharmacology of NX-2127, an Orally Bioavailable Degradable of Bruton's Tyrosine Kinase with Immunomodulatory Activity for the Treatment of Patients with B Cell Malignancies. *J. Med. Chem.* **2024**, *67* (4), 2321–2336.
- (14) Chen, C.; Yang, L. M. Ni(II)-(σ-Aryl) Complex: A Facile, Efficient Catalyst for Nickel-Catalyzed Carbon-Nitrogen Coupling Reactions. *J. Org. Chem.* **2007**, *72* (16), 6324–6327.
- (15) de Lange, B.; Lambers-Verstappen, M.; Schmieder-van de Vondervoort, L.; Sereinig, N.; de Rijk, R.; de Vries, A.; de Vries, J. Aromatic Amination of Aryl Bromides Catalysed by Copper/β-Diketone Catalysts: The Effect of Concentration. *Synlett* **2006**, 3105–3109.
- (16) Tao, Y.; Keene, N. F.; Wiglesworth, K. E.; Sitter, B.; McWilliams, J. C. Early Process Development of an Irreversible Epidermal Growth Factor Receptor (EGFR) T790 M Inhibitor. *Org. Process Res. Dev.* **2019**, *23* (3), 382–388.
- (17) Brusoe, A. T.; Hartwig, J. F. Palladium-Catalyzed Arylation of Fluoroalkylamines. *J. Am. Chem. Soc.* **2015**, *137* (26), 8460–8468.
- (18) Vasudevan, J.; Beard, R.; Huang, D.; Chandraratna, R. [54] A M INES SUBSTITUTED WITH A. US006093838, **2000**, July 25.
- (19) Carey, J. S.; Laffan, D.; Thomson, C.; Williams, M. T. Analysis of the Reactions Used for the Preparation of Drug Candidate Molecules. *Org. Biomol. Chem.* **2006**, *4* (12), 2337–2347.
- (20) Cooper, T. W. J.; Campbell, I. B.; MacDonald, S. J. F. Factors Determining the Selection of Organic Reactions by Medicinal Chemists and the Use of These Reactions in Arrays (Small Focused Libraries). *Angewandte Chemie - International Edition* **2010**, *49*, 8082–8091.
- (21) Roughley, S. D.; Jordan, A. M. The Medicinal Chemist's Toolbox: An Analysis of Reactions Used in the Pursuit of Drug Candidates. *J. Med. Chem.* **2011**, *54*, 3451–3479.
- (22) Brown, D. G.; Boström, J. Analysis of Past and Present Synthetic Methodologies on Medicinal Chemistry: Where Have All the New Reactions Gone? *J. Med. Chem.* **2016**, *59* (10), 4443–4458.
- (23) Dong, Z.; MacMillan, D. W. C. Metalla photoredox-Enabled Deoxygenative Arylation of Alcohols. *Nature* **2021**, *598* (7881), 451–456.
- (24) Zhang, H.; Ruiz-Castillo, P.; Schuppe, A. W.; Buchwald, S. L. Improved Process for the Palladium-Catalyzed C-O Cross-Coupling of Secondary Alcohols. *Org. Lett.* **2020**, *22* (14), 5369–5374.
- (25) Young, I. S.; Simmons, E. M.; Fenster, M. D. B.; Zhu, J. J.; Katipally, K. R. Palladium-Catalyzed C-O Coupling of a Sterically Hindered Secondary Alcohol with an Aryl Bromide and Significant Purity Upgrade in the API Step. *Org. Process Res. Dev.* **2018**, *22* (5), 585–594.
- (26) Chen, Z.; Jiang, Y.; Zhang, L.; Guo, Y.; Ma, D. Oxalic Diamides and *Tert*-Butoxide: Two Types of Ligands Enabling Practical Access to Alkyl Aryl Ethers via Cu-Catalyzed Coupling Reaction. *J. Am. Chem. Soc.* **2019**, *141* (8), 3541–3549.
- (27) Gowrisankar, S.; Sergeev, A. G.; Anbarasan, P.; Spannenberg, A.; Neumann, H.; Beller, M. A General and Efficient Catalyst for Palladium-Catalyzed C–O Coupling Reactions of Aryl Halides with Primary Alcohols. *J. Am. Chem. Soc.* **2010**, *132* (33), 11592–11598.

- (28) Zhang, H.; Ruiz-Castillo, P.; Buchwald, S. L. Palladium-Catalyzed C-O Cross-Coupling of Primary Alcohols. *Org. Lett.* **2018**, *20* (6), 1580–1583.
- (29) Ahneman, D. T.; Estrada, J. G.; Lin, S.; Dreher, S. D.; Doyle, A. G. Predicting Reaction Performance in C-N Cross-Coupling Using Machine Learning. *Science* **2018**, *360* (6385), 186–190.
- (30) Kwon, Y.; Lee, D.; Kim, J. W.; Choi, Y.-S.; Kim, S. Exploring Optimal Reaction Conditions Guided by Graph Neural Networks and Bayesian Optimization. *ACS Omega* **2022**, *7* (49), 44939–44950.
- (31) Saeibi, M.; Nan, B.; Herr, J. E.; Wahlers, J.; Guo, Z.; Zurański, A. M.; Kogej, T.; Norrby, P.-O.; Doyle, A. G.; Chawla, N. V.; Wiest, O. On the Use of Real-World Datasets for Reaction Yield Prediction. *Chem. Sci.* **2023**, *14* (19), 4997–5005.
- (32) Shields, B. J.; Stevens, J.; Li, J.; Parasram, M.; Damani, F.; Alvarado, J. I. M.; Janey, J. M.; Adams, R. P.; Doyle, A. G. Bayesian Reaction Optimization as a Tool for Chemical Synthesis. *Nature* **2021**, *590* (7844), 89–96.
- (33) Torres, J. A. G.; Lau, S. H.; Anchuri, P.; Stevens, J. M.; Tabora, J. E.; Li, J.; Borovika, A.; Adams, R. P.; Doyle, A. G. A Multi-Objective Active Learning Platform and Web App for Reaction Optimization. *J. Am. Chem. Soc.* **2022**, *144* (43), 19999–20007.
- (34) Fu, Z.; Li, X.; Wang, Z.; Li, Z.; Liu, X.; Wu, X.; Zhao, J.; Ding, X.; Wan, X.; Zhong, F.; Wang, D.; Luo, X.; Chen, K.; Liu, H.; Wang, J.; Jiang, H.; Zheng, M. Optimizing Chemical Reaction Conditions Using Deep Learning: A Case Study for the Suzuki-Miyaura Cross-Coupling Reaction. *Org. Chem. Front.* **2020**, *7*, 2269–2277.
- (35) Douglas, J. J.; Campbell, A. D.; Buttar, D.; Fairley, G.; Johansson, M. J.; McIntyre, A. C.; Metrano, A. J.; Morales, R. S.; Munday, R. H.; Nguyen, T. V. Q.; Staniland, S.; Tavanti, M.; Weis, E.; Yates, S. D.; Zhang, Z. The Implementation and Impact of Chemical High-Throughput Experimentation at AstraZeneca. *ACS Catal.* **2025**, *15* (7), 5229–5256.
- (36) Takáts, Z.; Wiseman, J. M.; Gologan, B.; Cooks, R. G. Mass Spectrometry Sampling Under Ambient Conditions with Desorption Electrospray Ionization. *Science* **2004**, *306* (5695), 471–473.
- (37) Dirico, K. J.; Hua, W.; Liu, C.; Tucker, J. W.; Ratnayake, A. S.; Flanagan, M. E.; Troutman, M. D.; Noe, M. C.; Zhang, H. Ultra-High-Throughput Acoustic Droplet Ejection-Open Port Interface-Mass Spectrometry for Parallel Medicinal Chemistry. *ACS Med. Chem. Lett.* **2020**, *11* (6), 1101–1110.